

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 33

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 4 from:</i> • Designation of protected areas (SSSIs, MCZs) • Protection of wild birds and their nests (except some pests / game birds) • Protection of named mammals such as otters, dormice, red squirrels, badgers and their setts • Uprooting wild plants generally illegal • Bats must not be disturbed; roof timber treatments must be non-toxic to bats	4
01.2	• Appendix I: species threatened with extinction - all international (commercial) trade banned, except movement for conservation breeding • Examples: great apes, big cats, rhinos, blue whale • Appendix II: species that may become threatened if trade is not closely controlled - trade permitted under licence from countries where the species is well protected • Examples: Honduras mahogany, hippopotamus, green iguana, great white / basking shark, Venus fly trap	4
01.3	• Collection may be hard to police in remote areas / poor countries, but if the animals or products cannot be exported / imported there is no market • Removing the profit removes the incentive to collect	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30-31; AQA spec 3.1.2.3.2.	

Question	Marking guidance	Marks
02.1	• Normal diagonal mesh closes as the net fills / stretches so small fish cannot escape; square-mesh panels do not close so juveniles escape to grow and breed; TEDs let turtles out; larger mesh sizes generally • Minimum size lets fish reach breeding age before capture (eg UK lobster minimum carapace above breeding size, cod 35 cm); maximum size / V-notching egg-carrying females protects large, highly fecund breeding adults (Hebridean lobsters over 145 mm, Nassau grouper) • NTZs (Lundy, Lamlash Bay, Chagos, Great Barrier Reef) protect breeding populations and habitats; surplus young spill over and increase surrounding catches; damaged communities recover • Closed seasons, ideally in the breeding season, let fish grow and breed; fishermen may oppose because shoaling breeders are easy to catch	8
02.2	• Pelagic species form single-species shoals so fishing stops when the quota is reached • In mixed fisheries many species are caught together; once one species' quota is reached it must be discarded (dead) while fishing for others continues	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.	
03.1	• Trees (hazel, oak) are cut to ground level every 3-30 years and allowed to regrow from the stump • About 0.5 ha cut each year on a rotation (eg 8 years) so the wood is a patchwork of areas of different ages • Each age has its own community: newly cut areas get high light for wildflowers and butterflies; older areas give dense cover for nesting birds / dormice • Pollarding cuts above browsing height so deer / livestock cannot eat the regrowth	4
03.2	• Wolves (top predator, exterminated by 1926) reduced deer populations and changed their behaviour • Fewer young trees eaten so trees and riverside vegetation recovered • More trees allowed beaver populations to increase; beaver dams increased wetland area • Wetland species (birds, fish, invertebrates) became more common; overall biodiversity rose, though wolves cause problems for livestock farmers	4
03.3	• Undesirable species (introduced predators / competitors, or natural succession species such as scrub, conifers in the New Forest heathland) reduce survival of the target species • Removal restores suitable conditions (eg heathland for sand lizards and smooth snakes) or reduces predation / competition	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3-3.1.3.4.	
04.1	• Silver carp eat phytoplankton, bighead carp zooplankton, grass carp and tilapia vegetation, common carp bottom-feed on molluscs / insects, catfish or perch eat small fish • Species occupy different niches / trophic levels so they do not compete and all food sources are used • Bottom feeders re-suspend nutrients boosting plankton / plant growth • Predators control over-breeding species (tilapia) so remaining fish grow larger; total yield exceeds any single species	4

Question	Marking guidance	Marks
04.2	• IMTA: fed species (salmon, shrimp) alongside inorganic extractive species (seaweeds absorbing ammonia and phosphate) and organic extractive species (filter-feeding mussels eating waste particles / plankton) • Wastes of one species become nutrients for another so less pollution and extra harvests • Aquaponics: aquaculture water rich in nutrients / organic matter feeds hydroponic salad crops (lettuce, herbs, watercress) • Plants remove the nutrients that would cause deoxygenation and cleaned water returns to the fish (tilapia, carp, catfish); rice-fish paddies similar	4
04.3	• Bluefin tuna do not breed well in captivity so young fish are caught from the wild and fattened in sea cages • Production still depends on (and removes juveniles from) the wild population; carnivores need large fish inputs	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 357-358; AQA spec 3.5.2.3.	
05.1	• Plants close their stomata to reduce transpiration losses • Closed stomata stop the absorption of carbon dioxide so photosynthesis and growth stop • Less water for nutrient absorption / transport and cell turgidity; severe shortage dehydrates cells and inhibits biochemical reactions	3
05.2	• Anaerobic soil favours denitrifying bacteria and inhibits nitrifying bacteria so nitrate is lost faster than it forms; higher risk of fungal disease; root respiration inhibited • Drainage ditches or pipes • Deep ploughing; avoiding compaction by machinery / livestock; adding organic matter to encourage worms	4
05.3	• Groundwater often contains dissolved salts; when the water evaporates from the soil the salts are left behind and accumulate • High soil salinity causes osmotic dehydration of roots / crop death; washing out with extra water uses more water and salinises rivers	2
05.4	• Add crushed lime (calcium carbonate / hydroxide); low pH increases leaching, inhibits nutrient uptake and mobilises toxic aluminium and lead	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 308-310, 313; AQA spec 3.5.1.1.	
06.1	• Non-indigenous species (Sitka spruce from North America) chosen for fast growth and pest resistance; indigenous wildlife's needs for food, pollination, seed dispersal and habitat are not met • Single species: easy planting, thinning and felling over large areas; but each tree species supports its own community so few species overall • Close planting makes trees compete for light so they grow tall and straight (good timber) but shades out ground flora and the species that depend on them • Simple age structure: uniform size makes management easy but gives little variety of light, temperature, wind; no old trees or dead wood for invertebrates and woodpeckers • Mixed woodland: varied niches, stratification, dead wood, indigenous trees each with their own dependent species	6

Question	Marking guidance	Marks
06.2	• Trees are felled when income will be greatest - approaching full size when the growth rate starts to decline • Continuing to grow them ties up land and capital for little extra timber; natural forests keep old and dead trees	2
06.3	• Industrial society replaced species-specific uses (baskets, fencing, charcoal, tool handles) with large-scale uses: chipboard, plywood, pulp • Wood fuel replaced by fossil fuels / electricity; so traditional management declined and woods became uniform	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 365-367; AQA spec 3.5.3.2.	
07.1	• Chicken and salmon need far less feed per kg gained (1.2-1.8 vs 7.0) • Fish are ectotherms so no energy is used for thermoregulation and water supports their mass; poultry grow fast and are kept warm indoors with movement restricted • Cattle are large endotherms, use energy in movement, and lose energy in methane from cellulose digestion	3
07.2	• Control of movement (housing / restricted grazing) so less energy is used in muscle activity • Temperature control (heated / insulated buildings) so less energy is used in maintaining body temperature • Species selection: choose species with low FCRs (poultry, fish); high-energy easily digested diets; slaughter at the fastest growth stage	3
07.3	• Only about 10% of the energy at one trophic level is passed to the next; most is lost as heat in respiration / metabolism or in undigested material • Humans eating plants (primary consumers) receive more of the energy captured by photosynthesis than eating animals fed on those plants (secondary consumers) • Livestock better where land cannot grow crops (upland, semi-arid, steep, thin soils): grazing animals with symbiotic gut bacteria digest cellulose in grass that humans cannot • Omnivores (pigs) convert food wastes (dairy by-products, catering waste) into meat	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
08.1	• t-test • The data are measured (continuous) values and two means are being compared • The data are (assumed to be) normally distributed	3
08.2	• Use the Mann-Whitney U test instead (compares medians)	1
08.3	• There is a 2% probability that the difference occurred by random chance • As this is less than 0.05 the difference is statistically significant / the null hypothesis is rejected	2
08.4	• The ranges of values (mean plus or minus SD) do not overlap: woodland up to 1.6 and clearing down to 2.4 • About 68% of values lie within one SD of each mean so the two data sets are clearly separated	2

Question	Marking guidance	Marks
08.5	- Significance shows the difference is unlikely to be due to chance, not that a causal relationship exists - Another variable (eg topography / aspect of the sites) could cause both; a controlled experiment or further evidence is needed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
09.1	Environmental DNA (eDNA) analysis	1
09.2	- Transmitters are too heavy for small birds / their batteries would not last for a long-term study - Geolocator tags are small and only record time and light levels; latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded	2
09.3	- Each separate (regional) population has its own unique gene pool - DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Protected areas: Korup NP Cameroon (1260 km², 600 tree species, endemic <i>Ancistrocladus korupensis</i> with anti-HIV / antimalarial alkaloids); Tumucumaque NP Brazil (39 000 km², adjoining French Guiana reserve); Alto Maues; Rio Bravo Belize funded by charities, companies and ecotourism - Debt for nature swaps; sustainable exploitation: selective logging of high-value timber (mahogany) funding conservation; ITTO aims (largely unmet); CITES listing of timber species (Honduras mahogany Appendix II) - Ecotourism income (but visitor damage in Costa Rica, Malaysia); community projects and employment; carbon sequestration payments - Effectiveness: protected areas work where enforced but face illegal logging, mining, poaching, roads, conflict; drivers (population growth, demand for beef, soya, palm oil, timber, minerals) originate outside the forest; climate change cannot be controlled locally - Judgement: combination of large protected areas, economic alternatives for local people and reduced international demand needed; overall loss continues	9

Question	Marking guidance	Marks
10.2	Carbon sequestration reducing climate change; genetic resources / medicines; rainfall regulation; global biodiversity	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 56-57; AQA spec 3.1.2.3.4.	
11.1	Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) - Setting priorities: IUCN Red List criteria, EDGE, endemic, keystone, flagship species, population dispersal - Legislation and protocols: Wildlife and Countryside Act, designated areas (SSSI to MCZ), CITES Appendices, IWC, EU CFP, ITTO - successes and limits of enforcement - Captive breeding and release: selection, difficulties, techniques, release site criteria, hard / soft release, monitoring; seed banks - Habitat conservation: creation, design (area, shape, corridors, age structure, colonisation), abiotic and biotic management, plagioclimax maintenance, population control, reintroductions - Habitat case studies (woodland, rainforest, coral reefs, islands, mangroves, Antarctica) illustrating threats and conservation efforts - Monitoring and new technologies informing decisions - Evaluation: examples of success (Iberian lynx, Hawaiian goose, Scilly rats, Yellowstone) and failure (ITTO, tigers); cost, politics, climate change; judgement on the most effective combination Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-71; AQA spec 3.1.2.3.	25

Question	Marking guidance	Marks
11.2	Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Energy flow: autotrophs and heterotrophs; losses by reflection, transmission, respiration; 100 000 to 1 000 units; few food chains beyond four levels; plant food most productive; grazing animals where crops impossible; pigs on wastes • Energetics: intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels • Control of genetics: asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer • GM: Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape; advantages; risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity) • Sustainability links: energy dependence, trophic level of diet, feeding 12 billion people • Justified conclusion on the balance between productivity, efficiency and sustainability Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	25